

Does the J-space trade off against written chain of thought as problems get harder?

Homework Zero — Fall 2026. Model: Qwen3-4B. Code:
<https://github.com/avchadha/AI-Alignment>.

1 Hypothesis

Recorded before the main experiments (report/hypothesis.md, committed 2026-07-16 before any runs; see git history): **ablating the J-space impairs the multi-hop reasoning needed to solve mathematics problems; the impairment grows with problem difficulty; and explicit chain of thought (CoT) counteracts it, with the rescue increasing with the amount of reasoning externalized.** Rationale: the paper reads the J-space and written CoT as partially interchangeable scratchpads; if the J-space is the load-bearing internal half, multi-hop work should fail without it unless externalized, and harder problems — needing more intermediate state per step — should lean on it more. Operationalized predictions: (1) direct answers suffer more from J-space ablation than CoT answers, and far more than from a matched-norm random ablation; (2) the control–ablated gap shrinks monotonically with the enforced thinking budget; (3) relative loss grows with difficulty (GSM8K → MATH-500 → AIME, and within MATH-500 levels 1–5); (4) ablated models write longer CoT when permitted.

2 Experiment design

Three ablation arms crossed with five thinking budgets on fixed problem sets (paired comparisons).

Arms. *Control* (no intervention); *J-space ablation* — at every token position, across the workspace layer band, the residual stream’s projection onto the span of the $k = 10$ most strongly activated J-lens vectors is removed, never ablating tokens in the top-10 of a clean forward pass (the paper’s procedure); *random matched-norm* — a projection onto k fixed random directions in the same band, rescaled per position to remove exactly the norm the J-space ablation would remove there. The random arm distinguishes J-space-specific damage from an equally large generic perturbation.

External-CoT axis. Thinking budgets {0, 256, 512, 1024, 4096} tokens in Qwen3’s native thinking mode, enforced by the harness rather than by instructions: generations exceeding the budget receive a force-injected `</think>`, and in every arm the answer segment is teacher-forced to “The final answer is `\boxed{...}`” (48-token cap), because a pilot showed that instruction-based “answer directly” conditions leak CoT into the answer text.

Datasets. GSM8K (openai/gsm8k test, $n = 150$, seeded), MATH-500 (HuggingFaceH4/MATH-500, $n = 150$, stratified 30 per level 1–5), AIME 2024 — HuggingFaceH4/aime_2024, the 30 problems of the 2024 AIME I and II exams, 3 samples per problem — and a synthetic 1/2/3-hop chained-arithmetic probe (60 seeded problems), added after calibration revealed a floor effect on direct answering (§3). Each opposite outcome is visible by construction: no rescue leaves the ablated curves flat across budgets; generic damage makes the random arm track the J-space arm; a reversed difficulty interaction shows relative loss falling.

Metrics. Accuracy with 95% bootstrap confidence intervals over problems (AIME samples averaged within problem), paired McNemar tests, retained accuracy, CoT length, and budget-clip and extraction-failure rates.

Table 1: GSM8K and MATH-500 accuracy by arm and thinking budget.

budget	GSM8K			MATH-500		
	ctrl	J-abl	rand	ctrl	J-abl	rand
0	.153	.140	.100	.227	.220	.180
256	.380	.447	.353	.293	.347	.273
512	.753	.920	.767	.480	.573	.493
1024	.920	.947	.920	.740	.767	.700
4096	.967	.960	.973	.873	.847	.833

3 Experimental details

The intervention uses a pre-fitted community Jacobian lens for Qwen3-4B (neuronpedia/jacobian-lens, fitted on wikitext with the anthropics/jacobian-lens recipe); its readout reproduced the paper’s three-regime walkthrough example before acceptance. The official repository provides readout code only, so the ablation is implemented here (src/jspace/ablation.py): J-lens vectors are computed with the final RMSNorm folded into the unembedding, the joint projection onto the selected k -vector span is removed by QR decomposition, and the clean-pass top-10 exclusion runs a second, hook-free forward pass ($\approx 2\times$ cost). Vectors are ranked by raw lens logit; cosine ranking was also calibrated and agreed, but promoted low-norm junk tokens.

On 30 held-out calibration problems, control direct-answer accuracy is 16.7% versus 86.7% with 1,024 thinking tokens; the medium layer band (14–29 of 36) was frozen for producing the largest direct-answer drop (to 6.7%; tied with the heavy band under logit ranking) while SST-2 classification remained intact under every band and selection rule (89–90%, versus 89% control). The J-space-versus-random comparison was deliberately excluded from calibration criteria.

The main run comprises 6,750 generations (3 arms $\times$ 5 budgets $\times$ 450 items) on one H100 (≈ 13.5 GPU-hours), bf16, fixed seeds, resumable JSONL; answers are judged by balanced-brace `\boxed{}` extraction with math-verify equivalence for MATH-500. Reproduction commands and settings: README.md and configs/frozen.yaml.

4 Experimental results

All 6,750 generations completed with no extraction failures. Full tables: results/analysis_main.txt and results/extra_stats.txt; all five figures: report/figs/.

(a) No selective impairment at budget 0 (prediction 1 not supported). On GSM8K and MATH-500 the direct-answer baseline sits at the floor calibration predicted, and J-ablation moves it by roughly one point (McNemar $p = 0.80 / 1.00$). On the arithmetic probe, whose baseline has headroom (66.7%), J-ablation does impair direct answering (53.3%; $p = 0.0078$) with a clean hop gradient — 1-hop unaffected (100%), 2-hop 75% $\rightarrow$ 55%, 3-hop 25% $\rightarrow$ 5% — but the random arm lands in the same place (51.7%; 100/50/5% by hops; J-space versus random $p = 1.0$). The damage is multi-hop-specific but not J-space-specific.

(b) A J-space-specific accuracy gain at intermediate budgets. At budget 512 the J-ablated arm exceeds control by 16.7 points on GSM8K (92.0% vs 75.3%; McNemar 27 vs 2, $p < 10^{-5}$) and 9.3 points on MATH-500 (57.3% vs 48.0%; $p = 0.0043$), with the same sign at budget 256. Unlike the impairment in (a), this effect is specific to the J-space directions: the random arm tracks control (76.7% / 49.3% at 512), and paired J-space-versus-random tests give $p < 10^{-4}$ (GSM8K, 512), $p = 0.024$ (GSM8K, 256), and $p = 0.017$ (MATH-500, 512).

(c) The mechanism is shorter chain of thought (prediction 4 inverted). At the effectively unconstrained 4,096 budget, J-ablated

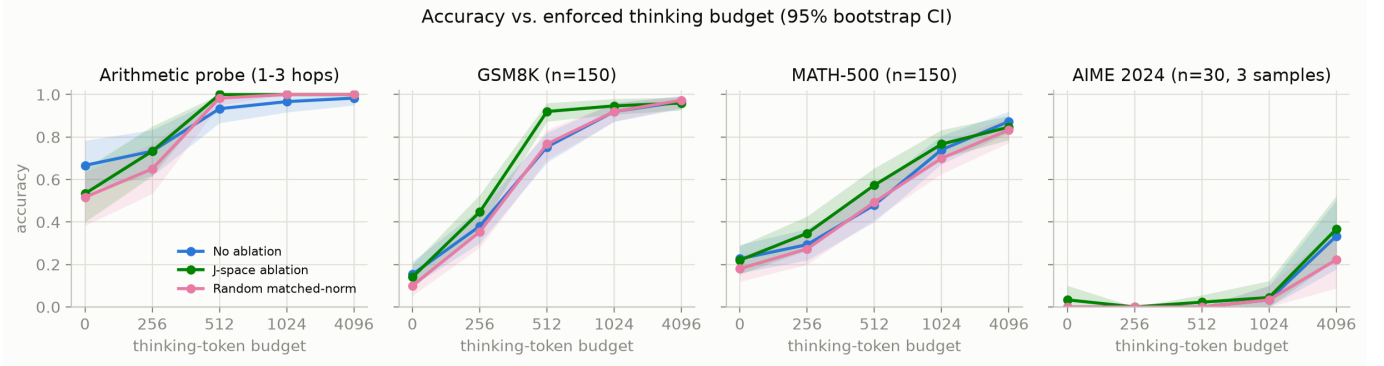


Figure 1: Accuracy versus enforced thinking budget by arm, with 95% bootstrap confidence intervals.

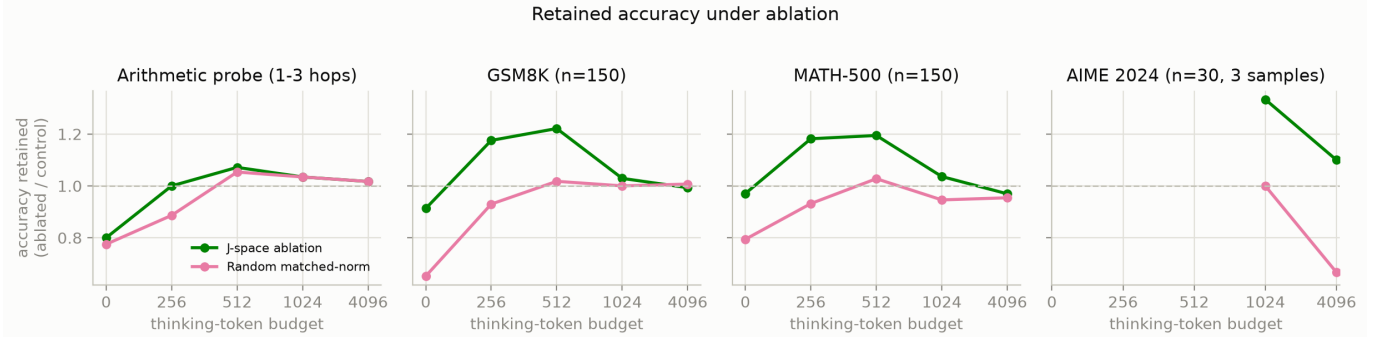


Figure 2: Retained accuracy (ablated/control) by thinking budget. The random arm hovers near 1.0 while the J-ablated arm exceeds it at intermediate budgets; AIME points below budget 1024 are undefined because control accuracy is zero there.

CoT is consistently shorter: 1,202 versus 1,703 mean thinking tokens on GSM8K (random: 1,642), 853 versus 1,049 on the arithmetic probe, 2,326 versus 2,657 on MATH-500 (length distributions: `report/figs/fig5_cot_length.png`). The J-ablated arm is correspondingly clipped less wherever clipping is common (GSM8K at 1024: 41% versus 69%), and the gain in (b) is concentrated exactly where control chains are truncated mid-reasoning.

(d) Dose-response and convergence (prediction 2, weak form).

Every arm rises steeply with budget and the arms converge by budgets 1024–4096 (GSM8K 96–97%; MATH-500 83–87%). CoT fully rescues the arithmetic impairment — the J-ablated arm reaches 100% at budget 512 (control 93.3%, random 98.3%; all arms $\geq 96.7\%$ by 1024) — but from damage that was never J-space-specific.

(e) No difficulty interaction in the predicted direction (prediction 3 not supported).

Within MATH-500 at budget 512 the J-ablation *advantage* appears at levels 1–4 (+.10, +.17, +.10, +.10) and vanishes at level 5 (.23 both; per-level curves: `report/figs/fig3_difficulty.png`). AIME is floored below budget 4096 (0–4%) and shows no significant arm differences at 4096 (control 33.3%, J-ablated 36.7%, random 22.2%); 89–94% of AIME chains still clip at 4,096 tokens, so that condition largely measures truncation.

5 Analysis of results

Of the four pre-registered predictions, only the weak form of prediction 2 (rescue by CoT, with dose-response) survives. Prediction 1’s selective impairment appears only where the direct-answer baseline is off the floor, and is matched by an equally large random perturbation; prediction 3’s difficulty interaction is absent; prediction 4 inverted. The one robustly J-space-specific effect is opposite in direction to the hypothesis: removing the top- k J-lens directions makes written CoT

shorter (median paired length ratio 0.76) at equal or better accuracy. A post-hoc transcript probe (`scripts/verify_reflection.py`; 20 GSM8K problems regenerated with full think-text, which the main run did not store) shows the omitted content is disproportionately *reflective*: J-ablated chains carry fewer reflection markers per thinking token (6.7 vs 9.9 per 1,000; “wait” 1.9 vs 3.2, “alternatively” 1.6 vs 3.9, “verify” 0.04 vs 0.25) — roughly half the double-checking and re-derivation loops per problem. This suggests the ablated directions in this band carry self-monitoring or verbalization content rather than load-bearing intermediate state; on this model the J-space–CoT relationship resembles modulation of how much the model narrates, not an internal scratchpad for which text substitutes.

Two qualifications: floor effects — when externalization is genuinely prevented, un-ablated Qwen3-4B scores 15–23% on GSM8K/MATH-500 and 0% on AIME, leaving little internal reasoning to ablate, a caution against extrapolating the paper’s headline result to small thinking-tuned models — and alternative explanations: the wikitext-fitted community lens may capture a shallower subspace than a paper-faithful lens, the random arm matches removed norm per position but not its location in the residual basis, and thinking-mode sampling adds variance. None of these plausibly explains the intermediate-budget gain, which replicates across two datasets with the random arm flat.

With more time or compute: a paper-faithful lens fit; k and layer-band sweeps with per-position delta-norm audits; clamping or patching individual J-vectors to isolate the CoT-shortening direction; faithfulness checks on the shortened chains; and replication on a larger model whose direct-answer baseline is off the floor without a synthetic probe.